Random Variables and Distributions

Probability and Statistics Foundations — Part 1

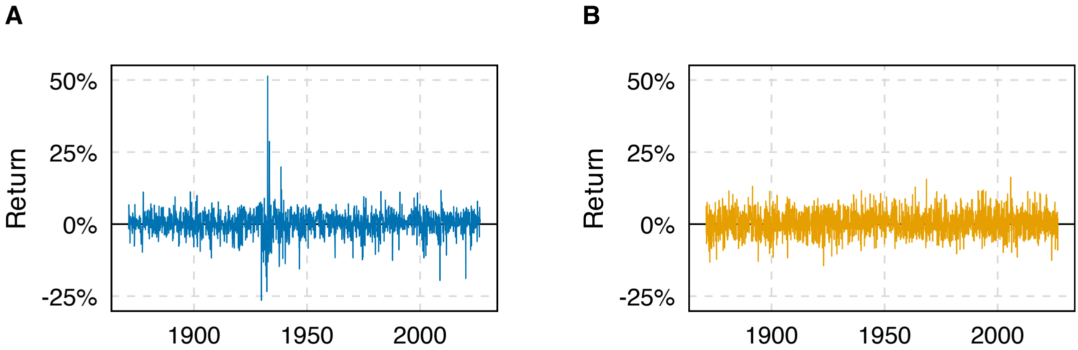
Instructor: Tudor Schlanger

Master's in Asset Management
Yale School of Management

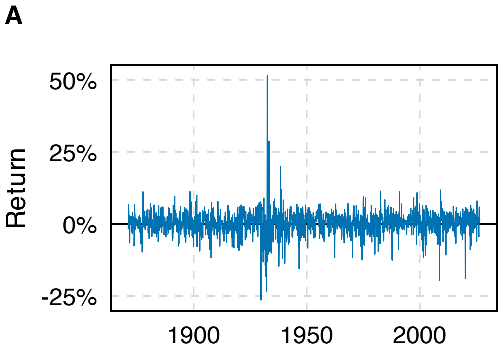
Objectives for this session

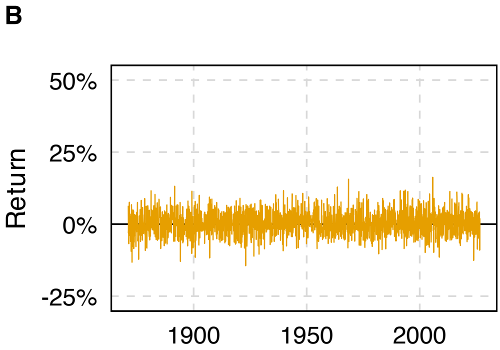
1. Expose you to ideas and terminology
2. Teach you how we map the real world to math objects
3. Teach you how to interpret math objects using real world examples

Random processes



Random processes

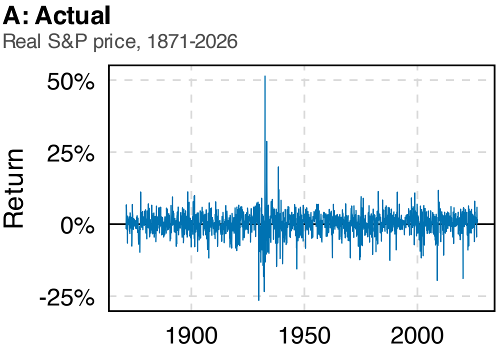


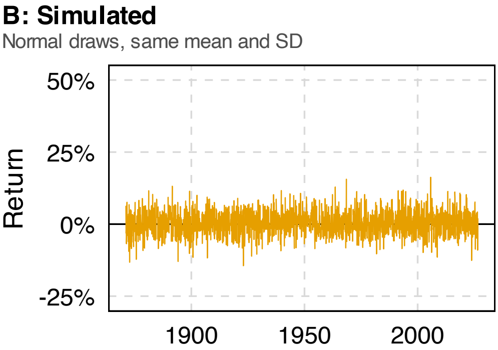


Question

Which one is actual return, which one is simulated?

Random processes



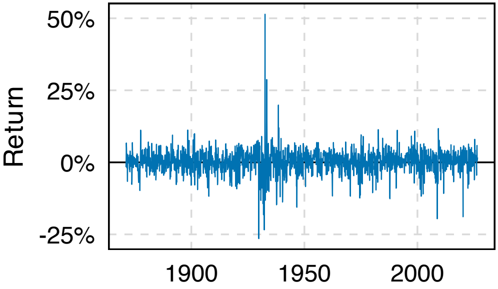


Question

Which one is actual return, which one is simulated?

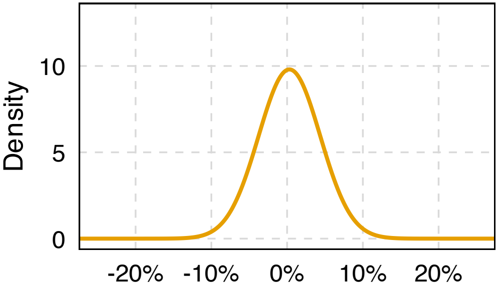
A: Actual

Real S&P price, 1871-2026



B: Simulated

The normal it is drawn from



Outline

1. Random variables and distributions
   How do we describe an uncertain outcome mathematically?

2. Population parameters and moments
   If we know the distribution, which summary numbers matter?

3. Statistical inference
   If we *don't* know the distribution, how do we learn it from data?

OUTLINE

1. **Random variables and distributions**
   How do we describe an uncertain outcome mathematically?
2. Population parameters and moments
   If we know the distribution, which summary numbers matter?
3. Statistical inference
   If we *don't* know the distribution, how do we learn it from data?

Start With a Data Generating Process

Data generating process (DGP)

Any mechanism that could, in principle, be **run indefinitely** and has a well-defined set of possible **outcomes**.

Before the DGP runs we do not know the outcome.
Afterwards we do.

Start With a Data Generating Process

Data generating process (DGP)

Any mechanism that could, in principle, be **run indefinitely** and has a well-defined set of possible **outcomes**.

Before the DGP runs we do not know the outcome. Afterwards we do.

Example 1:

- **DGP:** Flip a coin once
- **Outcome:** Heads

Example 2:

- **DGP:** Stock market trades for 1 day
- **Outcome:** Market price rises

Random Variables: From Outcomes to Numbers

A data generating process produces **outcomes**.
A random variable turns each outcome into a **number**.

Random Variables: From Outcomes to Numbers

A data generating process produces **outcomes**.
A random variable turns each outcome into a **number**.

Random variable

A random variable is a **function**.

$$X : \Omega \to \mathbb{R}$$

Random Variables: From Outcomes to Numbers

A data generating process produces **outcomes**.
A random variable turns each outcome into a **number**.

Random variable

A random variable is a **function**.

$$X : \Omega \to \mathbb{R}$$

Outcomes of the DGP (Ω)

Market falls

Market flat

Market rises

Random Variables: From Outcomes to Numbers

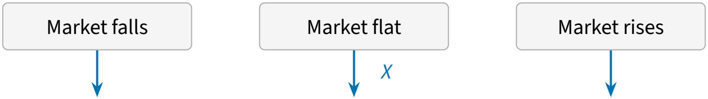
A data generating process produces **outcomes**.
A random variable turns each outcome into a **number**.

Random variable

A random variable is a **function**.

$$X : \Omega \to \mathbb{R}$$

Outcomes of the DGP (Ω)



Random Variables: From Outcomes to Numbers

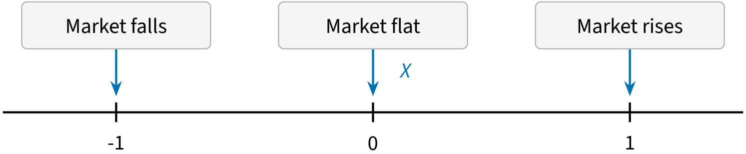
A data generating process produces **outcomes**.
A random variable turns each outcome into a **number**.

Random variable

A random variable is a **function**.

$$X : \Omega \to \mathbb{R}$$

Outcomes of the DGP (Ω)



The real line ($\mathbb{R}$)

Random Variables in Finance

Period	**DGP**	**Values X can take**
One trading day	Did the market rise?	$\{0, 1\}$
Two trading days	Number of up days	$\{0, 1, 2\}$
One week	Number of trades executed	$\{0, 1, 2, \ldots\}$
One month	% Return on the S&P 500	$[-100; \infty)$

Random Variables in Finance

Period	**DGP**	**Values X can take**
One trading day	Did the market rise?	$\{0, 1\}$
Two trading days	Number of up days	$\{0, 1, 2\}$
One week	Number of trades executed	$\{0, 1, 2, \ldots\}$
One month	% Return on the S&P 500	$[-100; \infty)$

> Look at the last column. Some lists are **discrete**, some are **continuous**.

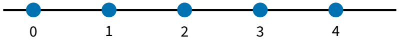
Two Kinds of Random Variables

DISCRETE RANDOM VARIABLE

Ω

Example: number of trades executed

X



gaps in between — nothing can land there

Two Kinds of Random Variables

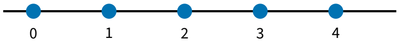
DISCRETE RANDOM VARIABLE

Ω

$\downarrow$

X

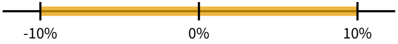
Example: number of trades executed



gaps in between — nothing can land there

CONTINUOUS RANDOM VARIABLE

Example: % return on the S&P 500



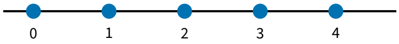
no gaps — every value in between is possible

Two Kinds of Random Variables

$$\Omega \;\downarrow\; X \;\downarrow\; \mathbb{P}(X = x)$$

DISCRETE RANDOM VARIABLE

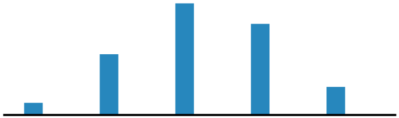
Example: number of trades executed



gaps in between — nothing can land there

Its distribution: a bar on each value

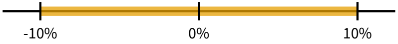
height $= \mathbb{P}(X = x)$



the five heights add to 1

CONTINUOUS RANDOM VARIABLE

Example: % return on the S&P 500



no gaps — every value in between is possible

Two Kinds of Random Variables

Ω

$\downarrow$

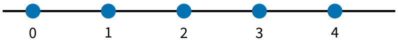
X

$\downarrow$

$\mathbb{P}(X = x)$

DISCRETE RANDOM VARIABLE

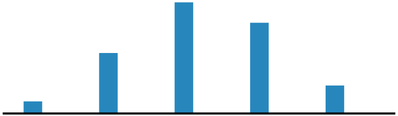
Example: number of trades executed



gaps in between — nothing can land there

Its distribution: a bar on each value

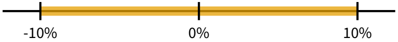
height $= \mathbb{P}(X = x)$



the five heights add to 1

CONTINUOUS RANDOM VARIABLE

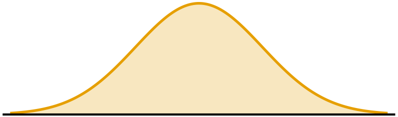
Example: % return on the S&P 500



no gaps — every value in between is possible

Its distribution: a curve over the range

area $= \mathbb{P}(a \leq X \leq b)$



the shaded area is 1

Notation: X Versus x

X — uppercase

The random variable itself

e.g. "X is next month's return."

Notation: X Versus x

X — uppercase

The random variable itself

e.g. "X is next month's return."

x — lowercase

A realized value.

e.g. "This month's return was $x = 2.3\%$."

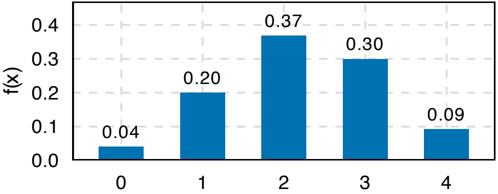
Discrete Random Variables

Probability mass function (pmf)

You can list the values $x_1, x_2, \ldots$

Its distribution is described by a **probability mass function**

$$f(x) = \mathbb{P}(X = x)$$



Some texts call this a pdf too; we say **pmf** for the discrete case.

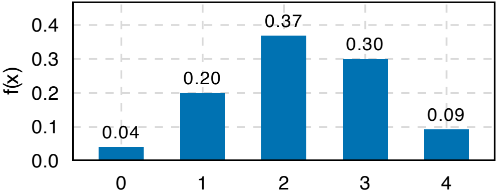
Discrete Random Variables

Probability mass function (pmf)

You can list the values $x_1, x_2, \ldots$

Its distribution is described by a **probability mass function**

$$f(x) = \mathbb{P}(X = x)$$



Some texts call this a pdf too; we say **pmf** for the discrete case.

Distributions: and their support

- Bernoulli $\{0, 1\}$
- Binomial $\{0, 1, 2, \ldots, N\}$
- Poisson $\{0, 1, 2, \ldots\}$

Discrete Random Variables

Probability mass function (pmf)

You can list the values $x_1, x_2, \ldots$

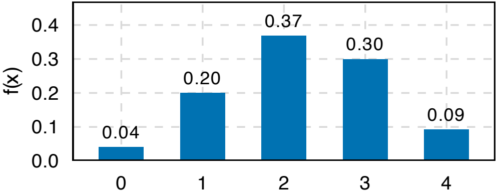
Its distribution is described by a **probability mass function**

$$f(x) = \mathbb{P}(X = x)$$

Some texts call this a pdf too; we say **pmf** for the discrete case.

Distributions: and their support

- Bernoulli $\{0, 1\}$
- Binomial $\{0, 1, 2, \ldots, N\}$
- Poisson $\{0, 1, 2, \ldots\}$



Some properties

1. $0 \le f(x) \le 1$ for every x
2. $\sum_x f(x) = 1$

Question

What is the probability that next month's S&P return is **exactly** 0.5%?

Continuous Random Variables

Question

What is the probability that next month's S&P return is **exactly** 0.5%?

Answer

Zero.

Not "very small" — exactly zero.

Question

What is the probability that next month's S&P return is **exactly** 0.5%?

Answer

Zero.

Not "very small" — exactly zero.

The values fill a range, so **no single value carries probability**: $\mathbb{P}(X = x) = 0$ for any x.

Question

What is the probability that next month's S&P return is **exactly** 0.5%?

Answer

Zero.

Not "very small" — exactly zero.

The values fill a range, so **no single value carries probability**: $\mathbb{P}(X = x) = 0$ for any x.

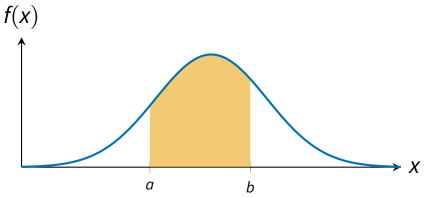
Why? A probability function must add up to 1.
If $\mathbb{P}(X = x) > 0$ for uncountably many values of x, the total could never stay finite — let alone equal 1.

Continuous Random Variables

Probability density function (pdf)

Probability attaches to a **range** $[a, b]$, through a **probability density function** $f(x)$:

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f(x)\,dx$$

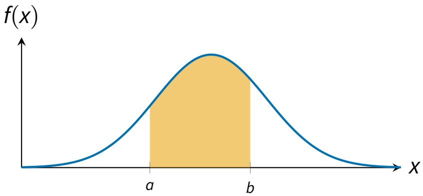


Continuous Random Variables

Probability density function (pdf)

Probability attaches to a **range** $[a, b]$, through a **probability density function** $f(x)$:

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f(x)\,dx$$



Distributions: and their support

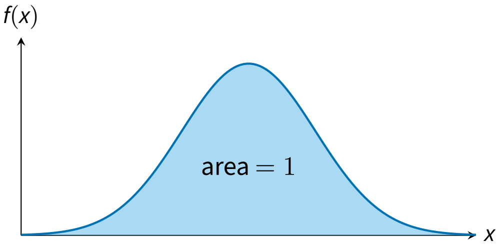
- Normal $(-\infty, \infty)$
- Lognormal $(0, \infty)$
- Student-t $(-\infty, \infty)$

Properties

Some properties

1. $f(x) \geq 0$ for every x
2. $\int_{-\infty}^{\infty} f(x)\, dx = 1$

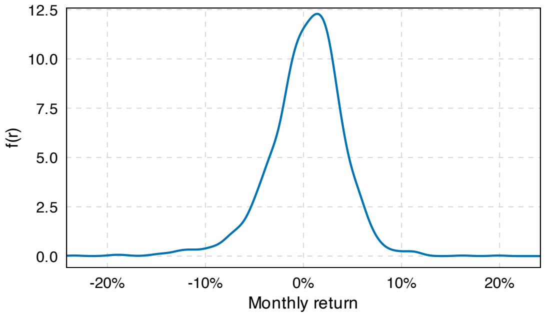
Note that $f(x)$ **can** exceed 1 — it is a *density*, not a probability.



Probabilities with Real Data

Question

What is the probability that the S&P 500 has a **losing** month?



Estimated density of real S&P monthly returns, 1871–2026.

Probabilities with Real Data

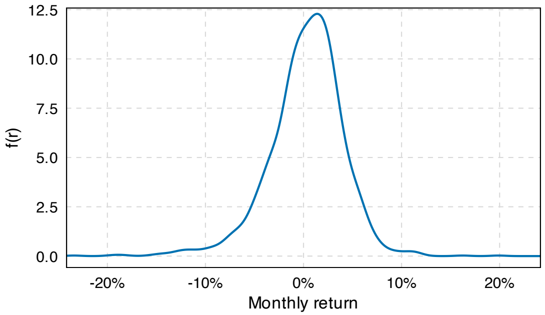
Question

What is the probability that the S&P 500 has a **losing** month?

In symbols, we want

$$\mathbb{P}(X < 0),$$

which is the area under the curve — the whole left tail.



Estimated density of real S&P monthly returns, 1871–2026.

Probabilities with Real Data

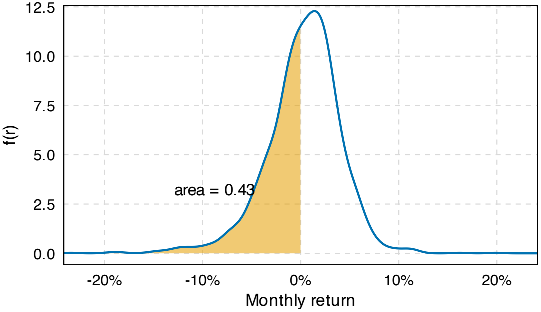
Question

What is the probability that the S&P 500 has a **losing** month?

In symbols, we want

$$\mathbb{P}(X < 0),$$

which is the area under the curve — the whole left tail.



Estimated density of real S&P monthly returns, 1871–2026.

Answer

About **0.43**

801 of the $1{,}867$ months since 1871.

From pdf to cdf

Cumulative Distribution Function (cdf)

For any number x,

$$F(x) = \mathbb{P}(X \leq x)$$

— all the area to the **left** of x.

1. $0 \leq F(x) \leq 1$ it is a probability

From pdf to cdf

Cumulative Distribution Function (cdf)

For any number x,

$$F(x) = \mathbb{P}(X \leq x)$$

— all the area to the **left** of x.

1. $0 \leq F(x) \leq 1$ it is a probability
2. $F(x) \to 0$ far to the left.
   $F(x) \to 1$ far to the right.

From pdf to cdf

Cumulative Distribution Function (cdf)

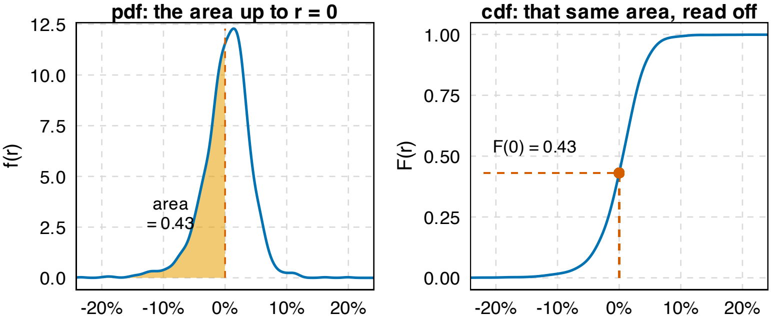
For any number x,

$$F(x) = \mathbb{P}(X \leq x)$$

— all the area to the **left** of x.

1. $0 \leq F(x) \leq 1$ it is a probability
2. $F(x) \to 0$ far to the left.
   $F(x) \to 1$ far to the right.
3. F never decreases. area only accumulates

The Same Idea, Two Pictures



More on CDF

Property 1 — the right tail

$$\mathbb{P}(X > c) = 1 - F(c)$$

Everything, minus what is to the left.

Property 2 — a middle slice

$$\mathbb{P}(a < X \leq b) = F(b) - F(a)$$

Left of b, minus left of a.

The Right Tail

Question

What is the probability that the S&P 500 monthly return is **positive**?

The Right Tail

Question

What is the probability that the S&P 500 monthly return is **positive**?

Everything, minus what is to the left of 0%:

$$\mathbb{P}(X > 0\%) = 1 - F(0\%)$$

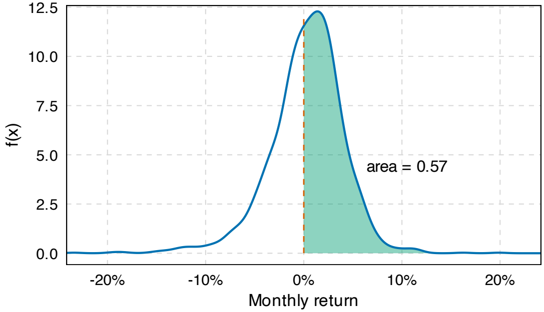
The Right Tail

Question

What is the probability that the S&P 500 monthly return is **positive**?

Everything, minus what is to the left of 0%:

$$\mathbb{P}(X > 0\%) = 1 - F(0\%)$$



Estimated density of real S&P monthly returns, 1871–2026. Shaded: everything above 0%.

The Right Tail

Question

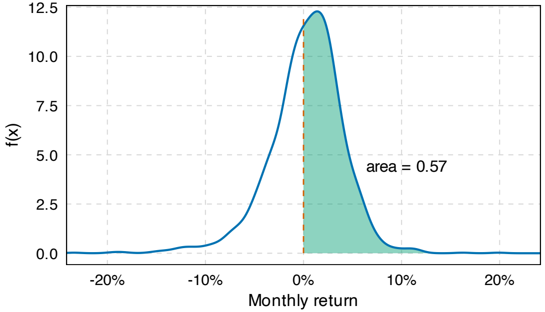
What is the probability that the S&P 500 monthly return is **positive**?

Everything, minus what is to the left of 0%:

$$\mathbb{P}(X > 0\%) = 1 - F(0\%)$$

Answer

$1 - 0.43 =$ **0.57**



Estimated density of real S&P monthly returns, 1871–2026. Shaded: everything above 0%.

A Middle Slice

Question

What is the probability that the S&P 500 monthly return falls between -5% and 0%?

A Middle Slice

Question

What is the probability that the S&P 500 monthly return falls between -5% and 0%?

Everything to the left of 0%, minus everything to the left of -5%:

$$\mathbb{P}(-5\% < X \leq 0\%) = F(0\%) - F(-5\%)$$

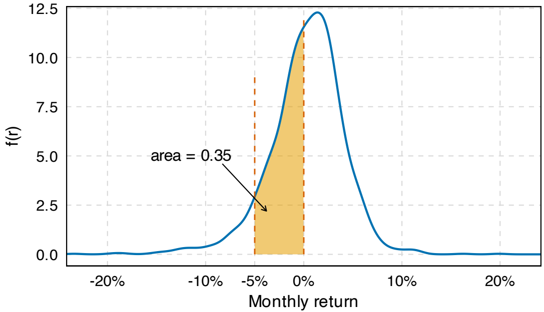
A Middle Slice

Question

What is the probability that the S&P 500 monthly return falls between -5% and 0%?

Everything to the left of 0%, minus everything to the left of -5%:

$$\mathbb{P}(-5\% < X \leq 0\%) = F(0\%) - F(-5\%)$$



Estimated density of real S&P monthly returns, 1871–2026. Shaded: -5% to 0%.

A Middle Slice

Question

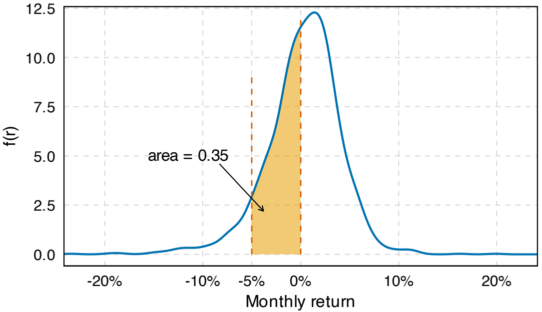
What is the probability that the S&P 500 monthly return falls between -5% and 0%?

Everything to the left of 0%, minus everything to the left of -5%:

$$\mathbb{P}(-5\% < X \leq 0\%) = F(0\%) - F(-5\%)$$

Answer

$0.431 - 0.078 = \mathbf{0.35}$



Estimated density of real S&P monthly returns, 1871–2026. Shaded: -5% to 0%.

The Limits of Empirical Distributions

Question

Are empirical distributions enough to understand reality?

- **Data hungry**. They require thousands of observations, which may be infeasible.
- **Sampling variability**. The distribution may change with the sample.
- **No algebra**. You cannot do math operations with empirical distributions.
- **No benchmark**. What should the distribution ideally look like?

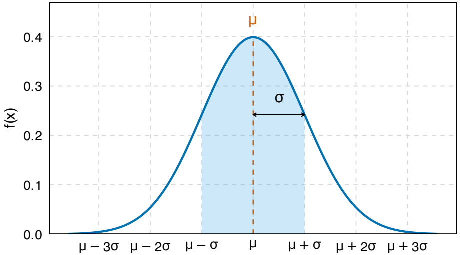
The Normal Distribution

Normal Distribution

We say that a random variable is distributed normal, $X \sim \mathrm{N}(\mu, \sigma^2)$, if its pdf is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R},$$

where μ sets the **centre** and σ the **spread**.



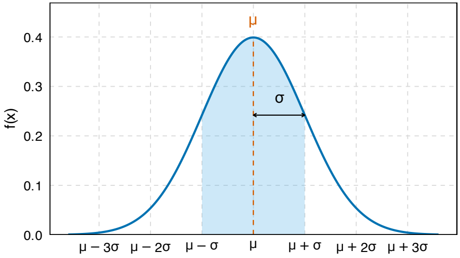
The Normal Distribution

Normal Distribution

We say that a random variable is distributed normal, $X \sim \mathrm{N}(\mu, \sigma^2)$, if its pdf is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R},$$

where μ sets the **centre** and σ the **spread**.



- **Two numbers, one curve.** Fix μ and σ and the whole distribution is known

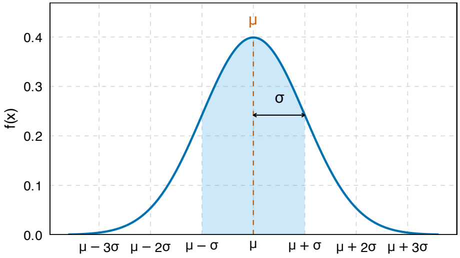
The Normal Distribution

Normal Distribution

We say that a random variable is distributed normal, $X \sim \mathrm{N}(\mu, \sigma^2)$, if its pdf is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R},$$

where μ sets the **centre** and σ the **spread**.



- **Two numbers, one curve.** Fix μ and σ and the whole distribution is known
- **Symmetric** around μ, so the centre is also the median: $\mathbb{P}(X > \mu) = \frac{1}{2}$.

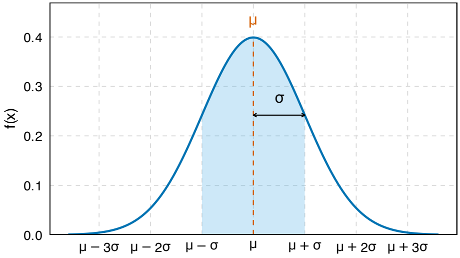
The Normal Distribution

Normal Distribution

We say that a random variable is distributed normal, $X \sim \mathrm{N}(\mu, \sigma^2)$, if its pdf is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R},$$

where μ sets the **centre** and σ the **spread**.



- **Two numbers, one curve.** Fix μ and σ and the whole distribution is known
- **Symmetric** around μ, so the centre is also the median: $\mathbb{P}(X > \mu) = \frac{1}{2}$.
- **Thin tails** the tails vanish at rate e^{-x^2}, so far-out values are very rare.

Is our empirical distribution Normal?

Question

Our monthly returns are continuous and look roughly symmetric. Does this curve describe them?

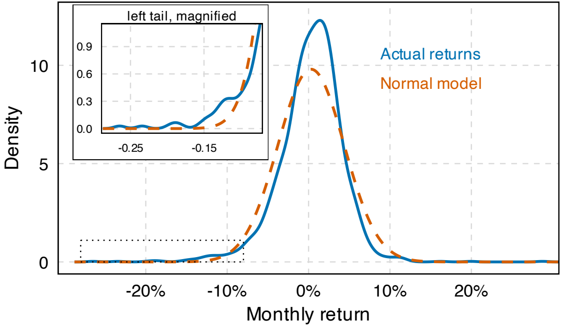
Is our empirical distribution Normal?

Question

Our monthly returns are continuous and look roughly symmetric. Does this curve describe them?

The empirical distribution of aggregate returns is not symmetric! We say

- It is skewed to the *left*, OR
- It has *negative* skew



Kernel density of $1{,}867$ monthly returns (blue) against a Normal with the *same* mean and standard deviation (red, dashed).

Fat Left Tail

	Probability the event occurs in a given month		
	Normal distribution	**Empirical distribution**	
Monthly **gain** better than 10%	0.86%	0.70%	about right
Monthly **gain** better than 15%	0.015%	0.21%	$14\times$ too rare

Fat Left Tail

	Probability the event occurs in a given month		
	Normal distribution	**Empirical distribution**	
Monthly **gain** better than 10%	0.86%	0.70%	about right
Monthly **gain** better than 15%	0.015%	0.21%	$14\times$ too rare
Monthly **loss** worse than 10%	0.57%	1.61%	$3\times$ too rare
Monthly **loss** worse than 15%	0.009%	0.32%	$38\times$ too rare

The model is symmetric about its *mean* of $+0.31\%$, not about zero: -10% sits $2.5\,\sigma$ out, $+10\%$ only $2.4\,\sigma$. That is why the two tails of the normal distribution carry different probabilities.

Fat Left Tail

	Probability the event occurs in a given month		
	Normal distribution	**Empirical distribution**	
Monthly **gain** better than 10%	0.86%	0.70%	about right
Monthly **gain** better than 15%	0.015%	0.21%	$14\times$ too rare
Monthly **loss** worse than 10%	0.57%	1.61%	$3\times$ too rare
Monthly **loss** worse than 15%	0.009%	0.32%	$38\times$ too rare

The model is symmetric about its *mean* of $+0.31\%$, not about zero: -10% sits $2.5\,\sigma$ out, $+10\%$ only $2.4\,\sigma$. That is why the two tails of the normal distribution carry different probabilities.

The model says a -15% month arrives roughly **once every 980 years**.

It has happened six times since 1871 — **once every 26 years**, most recently in March 2020.

Fat Left Tail

	Probability the event occurs in a given month		
	Normal distribution	**Empirical distribution**	
Monthly **gain** better than 10%	0.86%	0.70%	about right
Monthly **gain** better than 15%	0.015%	0.21%	$14\times$ too rare
Monthly **loss** worse than 10%	0.57%	1.61%	$3\times$ too rare
Monthly **loss** worse than 15%	0.009%	0.32%	$38\times$ too rare

The model is symmetric about its *mean* of $+0.31\%$, not about zero: -10% sits $2.5\,\sigma$ out, $+10\%$ only $2.4\,\sigma$. That is why the two tails of the normal distribution carry different probabilities.

The model says a -15% month arrives roughly **once every 980 years**.

It has happened six times since 1871 — **once every 26 years**, most recently in March 2020.

> Aggregate returns have **fat left tails**: extreme moves are far more common than a normal distribution implies.

Key Takeaways

Random variable	A **function** assigning a number to every outcome of the DGP. X is random, x is realized.
pmf, pdf	Where the probability sits. Discrete: a **pmf**, heights that sum to 1. Continuous: a **pdf**, a curve whose total area is 1.
cdf	$F(x) = \mathbb{P}(X \leq x)$.
Normal	$X \sim \mathrm{N}(\mu, \sigma^2)$. The bell curve, symmetric about μ and fixed entirely by a centre μ and a spread σ.
Fat left tail	Real returns are not normal. There is a higher probability of low returns than the Normal distribution would predict.

References

Shiller, Robert J. 2024. “U.S. Stock Markets 1871–Present and CAPE Ratio.” Online Data, https://shillerdata.com.

Bernoulli: One Day, Two Outcomes

Many questions in finance are yes/no:

- Did the market go up today?
- Did the borrower default?
- Did the fund beat its benchmark?

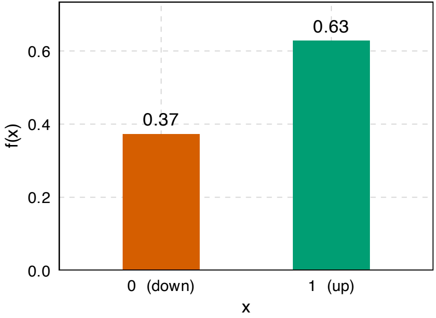
Bernoulli Random Variable

X takes only the values 0 and 1, with

$$\mathbb{P}(X = 1) = \theta, \qquad \mathbb{P}(X = 0) = 1 - \theta.$$

We write $X \sim \text{Bernoulli}(\theta)$.

One number, θ, describes it entirely.



Here $\theta = 0.63$: the share of up months for the US stock market, 1926–2023.

◂ Back

Binomial: Many Months

One month is a Bernoulli. But the interesting questions span **many** months:

Out of the next 3 months, how many will be up months?

Call that count X. It can be $0, 1, 2, 3$ — still discrete, but now with 4 possible values.

Two assumptions we need:

1. Each month has the **same** probability θ of being up.
2. Months are **independent** — this month tells us nothing about next month.

Both assumptions are **approximations**. Returns are not quite independent across months, and θ has surely not been one fixed number since 1926.

We adopt them anyway, because they buy us an exact answer. Knowing *which* assumptions you have made is half of statistics.

Back

Counting the Ways: a Small Case First

Before the general formula, take $n = 3$ months. In how many ways can we get exactly **2** up months?

Month 1 2 3:	U U U	U U D	U D U	D U U	U D D	D U D	D D U	D D D
	$X = 3$	$X = 2$	$X = 2$	$X = 2$	$X = 1$	$X = 1$	$X = 1$	$X = 0$

3 of the 8 sequences give $X = 2$. Each of those three has probability $\theta^2(1-\theta)$.

So $\mathbb{P}(X = 2) = 3 \cdot \theta^2(1-\theta)$.

The probability of *one* sequence is easy. The work is **counting the sequences**. That count is what the next formula supplies.

◂ Back

The Binomial Distribution

Binomial Distribution

If X counts the successes in n independent trials, each with probability θ, then

$$f(x) = \binom{n}{x} \theta^x (1-\theta)^{n-x}$$

for $x = 0, 1, \ldots, n$. We write $X \sim \text{Binomial}(n, \theta)$.

Our $n = 3, x = 2$ case:

$$\binom{3}{2} = 3 \quad \checkmark$$

The formula just automates the counting we did by hand.

Read it in three pieces:

- θ^x — x successes
- $(1-\theta)^{n-x}$ — the other $n - x$ are failures
- $\binom{n}{x}$ — how many orderings do that

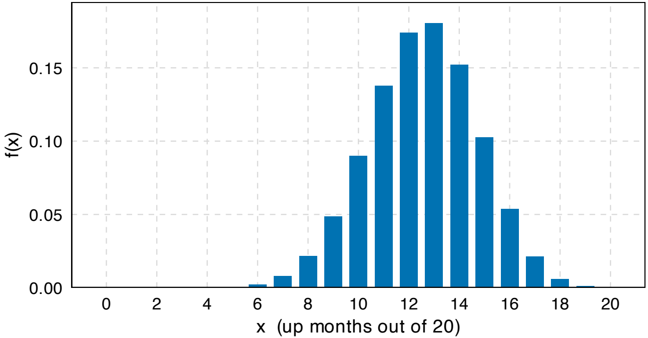
Back

Twenty Months

Set $n = 20$ and $\theta = 0.63$.

The whole distribution of "up months out of 20" is now pinned down by **two numbers**.

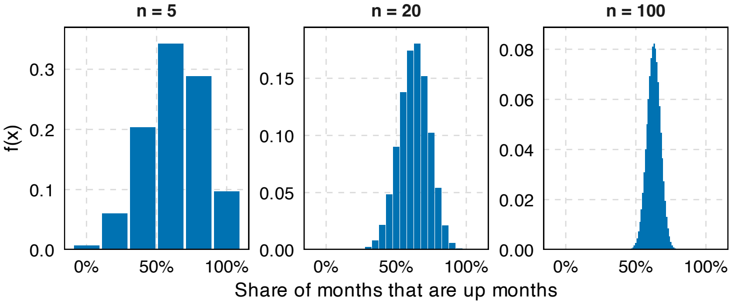
> The most likely single outcome is **13** up months.
>
> But it only happens about **18%** of the time.



◂ Back

What Happens as n Grows

Same $\theta = 0.63$, longer windows. The axis is the *share* of up months, so the panels are comparable.



Two things happen at once:

- The distribution **tightens** around 0.63.
- Its shape becomes smooth, symmetric, **bell-like**.

> Neither is a coincidence. The tightening is the **law of large numbers**; the bell shape is the **Central Limit Theorem**.

Poisson: Counts With No Upper Bound

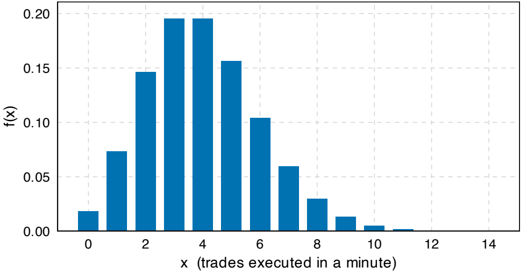
Poisson(λ)

Counts of events in a fixed window, when there is no natural **maximum**.

$$f(x) = \frac{e^{-\lambda}\lambda^x}{x!}, \quad x = 0, 1, 2, \ldots$$

Binomial counts successes out of n tries, so it stops at n. Trades arriving in a minute have no such ceiling — the support runs on forever.

$\mathbb{E}[X] = \mathrm{var}(X) = \lambda$. One parameter fixes both the level and the spread.



Poisson$(\lambda = 4)$. The bars continue to the right forever, each one smaller than the last.

◂ Back